

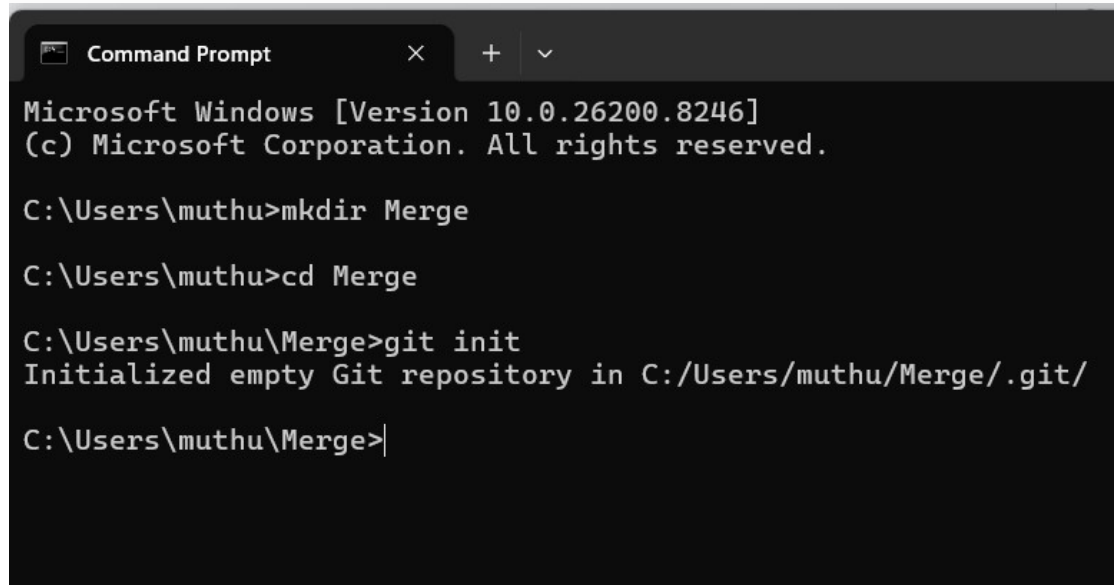
TITLE: - GITHUB MERGING EXERCISE

AIM: -

To perform Github merging exercise.

DESCRIPTION:

Start by creating a new repo.



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
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C:\Users\muthu>mkdir Merge

C:\Users\muthu>cd Merge

C:\Users\muthu\Merge>git init
Initialized empty Git repository in C:/Users/muthu/Merge/.git/

C:\Users\muthu\Merge>|
```

Make a file or two in the repo for you to work on.



```
C:\Users\muthu\Merge>echo "Hello" > greetings.txt

C:\Users\muthu\Merge>echo "# Git Merge Practice" > notes.txt

C:\Users\muthu\Merge>
```

If you need some inspiration...I'll be working in a file called `greetings.txt` It will contain greetings in different languages.

```
C:\Users\muthu\Merge>git add .

C:\Users\muthu\Merge>git commit -m "Initial Commit"
[master (root-commit) 5648ac0] Initial Commit
 2 files changed, 2 insertions(+)
 create mode 100644 greetings.txt
 create mode 100644 notes.txt

C:\Users\muthu\Merge>
```

Part 1: Fast Forward Merge

Your goal is to generate a fast forward merge. Demonstrate that you understand how FF merges work by creating one on your own!

Make a new branch. Do some work in the repo such that when you merge the new branch into master, it results in a fast forward merge. Merge that branch into master and see if you were right!

```
C:\Users\muthu\Merge>git checkout -b spanish-greetings
Switched to a new branch 'spanish-greetings'

C:\Users\muthu\Merge>echo "Hola" >> greetings.txt

C:\Users\muthu\Merge>git add greetings.txt

C:\Users\muthu\Merge>git commit -m "Add Spanish greeting"
[spanish-greetings e8ba0d6] Add Spanish greeting
 1 file changed, 1 insertion(+)

C:\Users\muthu\Merge>git checkout master
Switched to branch 'master'

C:\Users\muthu\Merge>git merge spanish-greetings
Updating 5648ac0..e8ba0d6
Fast-forward
 greetings.txt | 1 +
 1 file changed, 1 insertion(+)

C:\Users\muthu\Merge>
```

Part 2: Merge Commit (No Conflicts)

Your goal is to generate a merge commit with NO MERGE CONFLICTS.

Create a new branch. Make some changes to the repo such that when you merge the new branch into master, it results in a merge commit. The merge should not result in any conflicts. Merge that branch into master and see if you were right!

```
C:\Users\muthu\Merge>git checkout french-greetings
Switched to branch 'french-greetings'

C:\Users\muthu\Merge>echo "Bonjour" >> greetings.txt

C:\Users\muthu\Merge>git add greetings.txt

C:\Users\muthu\Merge>git commit -m "Add French greeting"
[french-greetings c4cc225] Add French greeting
1 file changed, 1 insertion(+)

C:\Users\muthu\Merge>git checkout master
Switched to branch 'master'

C:\Users\muthu\Merge>echo "Repository for merge exercises" >> notes.txt

C:\Users\muthu\Merge>git add notes.txt

C:\Users\muthu\Merge>git commit -m "Update notes"
[master 6f510dc] Update notes
1 file changed, 1 insertion(+)

C:\Users\muthu\Merge>git merge french-greetings
error: there was a problem with the editor 'vim'
Not committing merge; use 'git commit' to complete the merge.

C:\Users\muthu\Merge>git log --oneline --graph --all
* 6f510dc (HEAD -> master) Update notes
* 67fade1 Update notes
| * c4cc225 (french-greetings) Add French greeting
| * 9758c10 Add French greeting
|/
* e8ba0d6 (spanish-greetings) Add Spanish greeting
* 5648ac0 Initial Commit

C:\Users\muthu\Merge>
```



Part 3: Conflicts!

Your goal is to generate merge a conflict!

Create a new branch. Make some changes to the repo such that when you merge the new branch into the master branch, it results in a merge conflict. Merge that branch into master and see if you were right! Resolve the conflict!

```

C:\Users\muthu\Merge>git branch
french-greetings
german-greetings
* master
spanish-greetings

C:\Users\muthu\Merge>git branch
french-greetings
german-greetings
* master
spanish-greetings

C:\Users\muthu\Merge>git log --oneline --graph --all
* 5fa9008 (HEAD -> master) Change greeting to Italian
| * 37f7fec (german-greetings) Change greeting to German
|/
* 6f510dc Update notes
* 67fade1 Update notes
| * c4cc225 (french-greetings) Add French greeting
| * 9758c10 Add French greeting
|/
* e8ba0d6 (spanish-greetings) Add Spanish greeting
* 5648ac0 Initial Commit

C:\Users\muthu\Merge>git status
On branch master
All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

Changes to be committed:
  modified:   greetings.txt
  modified:   notes.txt

C:\Users\muthu\Merge>

```